# Problem Set 2

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## 1 Sources:

https://math.stackexchange.com/questions/2392992/matrix-condition-number-and-loss-of-accuracy
https://math.stackexchange.com/questions/675474/what-is-the-practical-impact-of-a-matrixs-condition-nur
~:text=In%20practice%2C%20a%20matrix%20with,precision%2C%20or%20arbitrary%20precision%20arithmetic
https://www.mathworks.com/help/matlab/ref/lu.html https://math.emory.edu/~lchen41/teaching/
2020_Fall/Section_8-3.pdf

## 2 Exercise 1

Using magic(n), we produce matrices with entries whose sum of elements by rows, columns, or diagonals is identical. Observing the plot, we can see that the actual CPU time is very close to the computation time $t_n$. I am not sure I can suspect which exact method Matlab uses to compute the determinant, but it was discussed in class that it uses LU decomposition. We know for a fact that it cannot be using co-factor expansion, since its complexity would be *O(n!)*. This means that for a large matrix, for example for n=100, it would take roughly $9.33 \times 10^{157}$ operations to compute the determinant. Since for n=100:100:1000, it took less than a minute to compute, the complexity has to be polynomial and not factorial.

## 3 Exercise 2

(a) $A = \begin{bmatrix} 201 & 199 \\ 199 & 201 \end{bmatrix}$.

We know that the spectral radius of matrix A is the maximum absolute value of the eigenvalues. In order to find the eigenvalues, we solve the characteristic equation $\det(A - \lambda I) = 0$:

$$\det\left(\begin{bmatrix} 201 & 199 \\ 199 & 201 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right) = 0$$

$$\det\begin{bmatrix} 201-\lambda & 199 \\ 199 & 201-\lambda \end{bmatrix} = 0$$

$$(201-\lambda)^2 - 199^2 = 0$$

$$((201-\lambda) - 199)((201-\lambda) + 199) = 0$$

$$(2-\lambda)(400-\lambda) = 0$$

So, our eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = 400$. Spectral radius of A:

$$\rho(A) = \max(|\lambda_1|, |\lambda_2|)$$

$$\rho(A) = \lambda_2 = 400$$

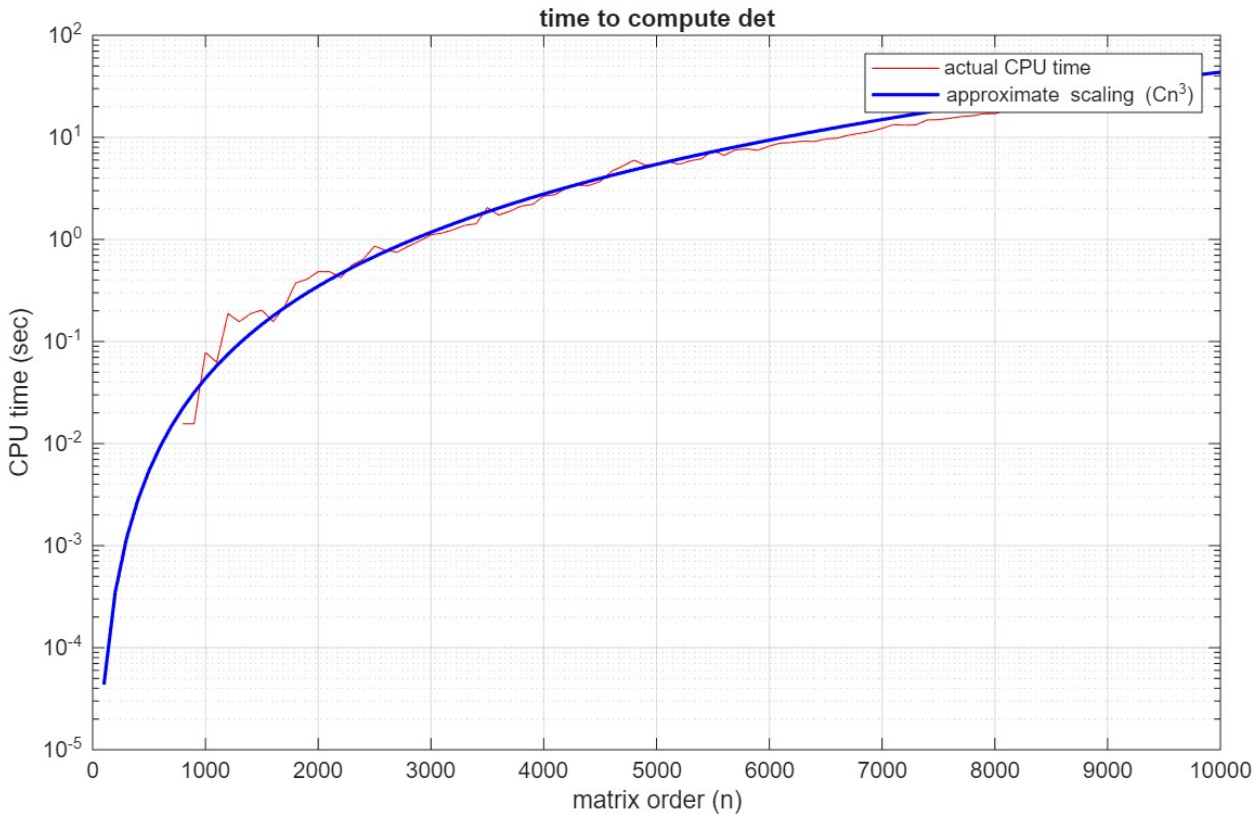


Figure 1: Determinant-Time

(b) To compute the condition number here, we will use the infinity norm. The condition number is the number defined as:

$$K_\infty(A) = \|A\|_\infty \|A^{-1}\|_\infty$$

We know that the infinity norm is the maximum absolute of the row sum (from both rows):

$$\|A\|_\infty = max(|201 + 199|, |199 + 201|) = max(|400|, |400|) = 400$$

To find the inverse of A, we first compute the determinant:

$$\det(A) = (201 * 201) - (199 * 199) = 800$$

We know for a 2x2 matrix the formula to find the inverse is:

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$A^{-1} = \frac{1}{800} \begin{bmatrix} 201 & -199 \\ -199 & 201 \end{bmatrix} = \begin{bmatrix} \frac{201}{800} & -\frac{199}{800} \\ -\frac{199}{800} & \frac{201}{800} \end{bmatrix}$$

$$\|A^{-1}\|_\infty = max((|\frac{201}{800}| + |-\frac{199}{800}|), (|-\frac{199}{800}| + |\frac{201}{800}|)) = max(\frac{400}{800}, \frac{400}{800}) = 0.5$$

therefore, the condition number is:

$$K_\infty(A) = 400 \times 0.5 = 200$$

(c) This is the approximate solution after some numerical method has been applied $\hat{x} = [-0.5, 0.5]^T$. In order to find a bound for the relative error, we must compute the residual vector $r = A\hat{x} - b$:

$$A\hat{x} = \begin{bmatrix} 201 & 199 \\ 199 & 201 \end{bmatrix} \begin{bmatrix} -0.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} -100.5 + 99.5 \\ -99.5 + 100.5 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$A\hat{x} - b = \begin{bmatrix} -1 \\ 1 \end{bmatrix} - \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

So our residual is $r = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. From our notes we know that the error bound is:

$$\frac{||e||}{||x||} \leq K(A) \cdot \frac{||r||}{||b||}$$

Since we used the infinity norms for the previous calculations, we will find $\|r\|_\infty$ and $\|b\|_\infty$: $\|r\|_\infty = 1$ $\|b\|_\infty = 2$ Substituting back to the error bound:

$$\frac{||e||}{||x||} \leq K(A) \cdot \frac{||r||}{||b||}$$

$$\frac{||e||}{||x||} \leq 200 \cdot \frac{1}{2} = 100$$

Our relative error bound is less than 100.

(d) To compare the relative error to the approximate error, we must solve the original system first.

$$Ax = b \implies$$

$$\begin{bmatrix} 201 & 199 \\ 199 & 201 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$$

$$201x_1 + 199x_2 = -2$$

$$199x_1 + 201x_2 = 2$$

$$(201 - 199)x_1 + (199 - 201)x_2 = -2 - 2$$

$$2x_1 - 2x_2 = -4$$

$$x_1 - x_2 = -2 (1)$$

$$(201 + 199)x_1 + (199 + 201)x_2 = -2 + 2$$

$$400x_1 + 400x_2 = 0$$

$$x_1 + x_2 = 0 \implies x_1 = -x_2$$

Substitute in (1):

$$(-x_2) - x_2 = -2 \implies -2x_2 = -2 \implies x_2 = 1$$

Therefore,

$$x_1 = -1$$

So, our exact solution is

$$x = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Relative error

$$\frac{||e||_\infty}{||x||_\infty} = \frac{||x - \hat{x}||_\infty}{||x||_\infty} = \frac{0.5}{1} = 0.5$$

The theoretical bound is much larger than the actual error of 0.5. This implies that the theoretical inequality $\frac{||e||}{||x||} \leq$K$(A) \cdot \frac{||r||}{||b||}$ provides a "worst-case scenario" bound. It guaranties that it won't be above 100, but in this case, the actual error is only 0.5.

(e) In general, when can you expect the residual r to provide a tight bound on the error? In general, the residual r provides a tight bound on the error when our condition number for the matrix is significantly small. We've previously established that the condition number is a measure of the sensitivity of the system we are looking at. The larger the condition number, the higher the chance our relative error is large. So we can expect the residual r to provide a tight bound on the error as long as the matrix is well-conditioned (small condition number).

# 4 Exercise 3

(a) Compute the LU factorization using Gaussian Elimination and show that L has $l_{32} = -\frac{1}{\epsilon}$

$$A = \begin{bmatrix} 2 & -2 & 0 \\ \epsilon - 2 & 2 & 0 \\ 0 & -1 & 3 \end{bmatrix} \xrightarrow{R_2 \to R_2 - \frac{\epsilon - 2}{2} R1} \begin{pmatrix} 2 & -2 & 0 \\ 0 & \epsilon & 0 \\ 0 & -1 & 3 \end{pmatrix} \xrightarrow{R_3 \to R_3 + \frac{1}{\epsilon} R_2} U = \begin{pmatrix} 2 & -2 & 0 \\ 0 & \epsilon & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

U will be the upper triangular matrix we get from applying Gaussian elimination to the original matrix A. My L entries $l_{21}$, $l_{31}$, and $l_{32}$ should be the scalars we used during row operations:

$$L = \begin{bmatrix} 2 & 0 & 0 \\ \frac{\epsilon - 2}{2} & \epsilon & 0 \\ 0 & \frac{-1}{\epsilon} & 0 \end{bmatrix}$$

L has $l_{32} = -\frac{1}{\epsilon}$.

(b) Compute the LU factorization using Crout's algorithm and show that L has $l_{22} = \epsilon$ in this case.
Using Crout's algorithm:
1st column of L:

$$l_{11} = a_{11} = 2$$

$$l_{21} = a_{21} = \epsilon - 2$$

$$l_{31} = a_{31} = 0$$

1st row of U:

$$uij = \frac{a_{ij}}{l_{11}}$$

$$u_{11} = 1$$

$$u_{12} = \frac{a_{12}}{l_{11}} = \frac{-2}{2} = -1$$

$$u_{13} = \frac{a_{13}}{l_{11}} = 0$$

2nd column of L:

$$l_{i2} = a_{i1} - l_{i1} \cdot u_{12}$$

$$l_{22} = a_{22} - l_{21} \cdot u_{12} = 2 - (\epsilon - 2) \cdot (-1) = \epsilon$$

$$l_{32} = a_{32} - l_{31} \cdot u_{12} = -1 - 0(-1) = -1$$

2nd row of U:

$$u_{2j} = \frac{1}{l_{22}} \cdot (a_{2j} - l_{21} \cdot u_{1j})$$

$$u_{22} = 1$$

$$u_{23} = \frac{1}{l_{22}} \cdot (a_{23} - l_{21} \cdot u_{13}) = 0$$

3rd column of L:

$$l_{33} = a_{33} - (l_{31} \cdot u_{13} + l_{32} \cdot u_{23}) = 3$$

L has $l_{22} = \epsilon$ with Crout's algorithm.

(c)The condition number of A is given by the formula:

$$K_\infty(A) = \|A\|_\infty \|A^{-1}\|_\infty$$

$$\|A\|_\infty = max((|2| + |-2|), (|\epsilon - 2| + |2|), (|-1| + |3|)) = max(4, 4 - \epsilon, 4) = 4$$

assuming that $\epsilon$ is very small.

$det(A) = 6\epsilon$

$$A^{-1} = \begin{bmatrix} \frac{1}{\epsilon} & \frac{1}{\epsilon} & 0 \\ \frac{2-\epsilon}{2\epsilon} & \frac{1}{\epsilon} & 0 \\ \frac{\epsilon-2}{6\epsilon} & \frac{1}{3\epsilon} & \frac{1}{3} \end{bmatrix}$$

$$\|A^{-1}\|_\infty = max((|\frac{1}{\epsilon}| + |\frac{1}{\epsilon}|), (|\frac{2-\epsilon}{2\epsilon}| + |\frac{1}{\epsilon}|), (|\frac{\epsilon-2}{6\epsilon}| + |\frac{1}{3\epsilon}| + |\frac{1}{3}|)) = max((\frac{2}{\epsilon}), (\frac{4-\epsilon}{2\epsilon}), (\frac{4+\epsilon}{6\epsilon})) = \frac{2}{\epsilon}$$

$$K(A) = 4 \cdot \frac{2}{\varepsilon} = \frac{8}{\varepsilon}$$

(d) Solve $A \cdot x = b$ for different values of $\epsilon$. After numerically implementing Crout's method and using the built in LU factorization in Matlab, we see that the errors depend on the values of $\epsilon$. Specifically, after k=2, the errors for both Crout's and the built-in method are the same. This happens because our $\epsilon$ values become very small, so both algorithms must reach the same precision limits. Especially with Crout's method, we divide by the value that includes $\epsilon$, so once it is almost close to 0, the round-off error takes over and there is no significant difference between these two methods. Also, for lower values of $\epsilon$, the conditioning number of A becomes significantly larger making the matrix highly unstable, therefore contributing to the fact that the errors are the same as we continue the iterations.

Below is the graph comparing the relative error with the different $\epsilon$ values:

We also notice that the built-in method "finishes" faster, however we know that both these methods require $O(n^3)$ number of operations. This is most likely due to the fact that it is more accurate than our Crout algorithm since the error is equal to exactly 0 faster, therefore it cannot be represented on our graph.

(e) After numerically implementing Crout's method and the built-in function for a different solution now, we get very different results, likely due to the nature of the log value in the solution as it is an irrational number. Now we see that after k=4, both errors become the same, due to the effect of the higher $\epsilon$ values on the condition number of the matrix. As $\epsilon$ shrinks by a factor of 10, the error grows by roughly a factor of 10, because the condition number is growing proportionally, making the matrix highly unstable. This is a direct result of the relationship between relative error, machine precision error, and condition number, which is described by:

$$error^{rel} \leq K(A) \cdot error^{mach}$$

. For example, at k=9, our condition number should be roughly $10^9$, while the machine precision error is $10^{-16}$, so our relative error should be approximately $10^{-7}$. This is verified by our results, since it is equal to $2.50e-07$.

Below is the graph comparing the relative error with the different $\epsilon$ values:

Also, we see that at k=3, both errors are equal to 0. Log values have to be rounded in order to be represented on a computer, therefore there must have been some rounding up and down during those calculations that cause the 0 to appear and the rest of the calculations to continue after k=3.

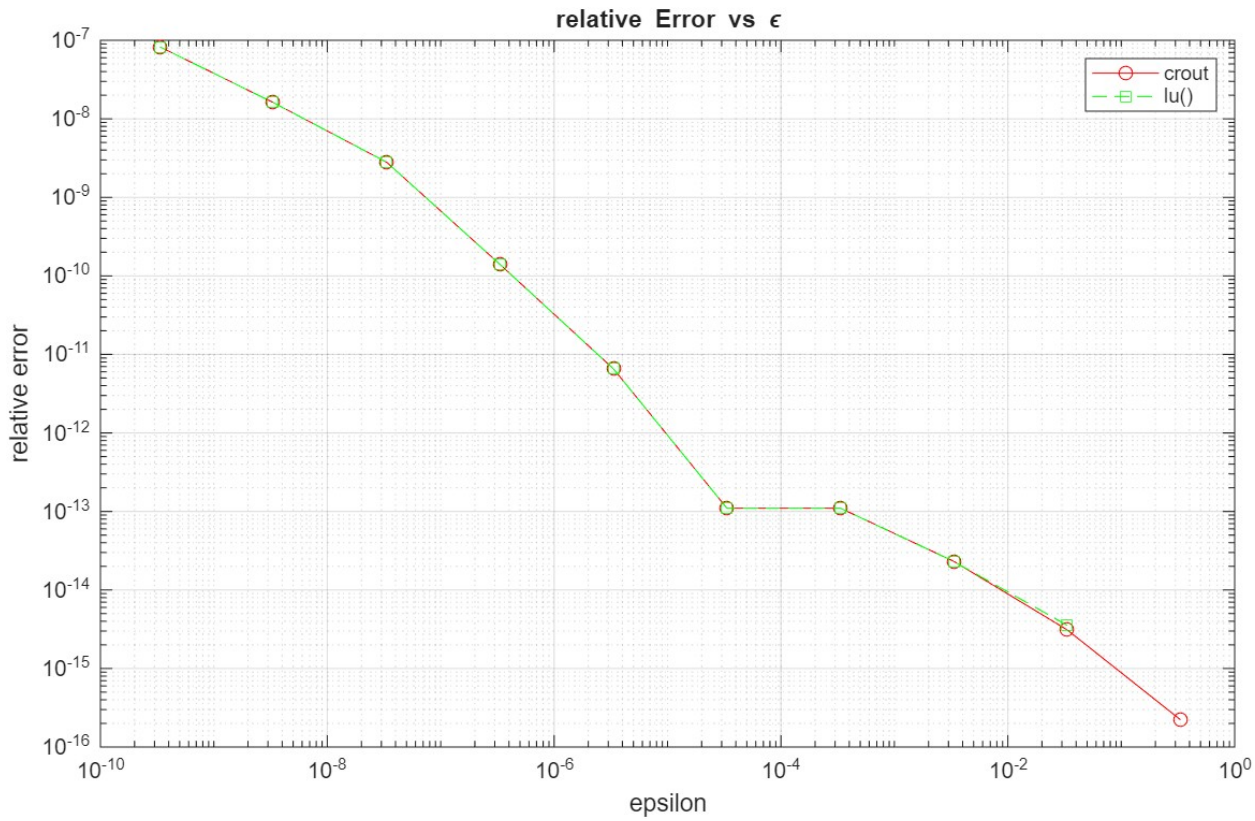


Figure 2: plot relative error vs epsilon

# 5 Exercise 4

LU decomposition with Crout's method (no pivoting): $A = \begin{bmatrix} 1 & 1+0.5\times 10^{-15} & 3 \\ 2 & 2 & 20 \\ 3 & 6 & 4 \end{bmatrix}$

1st column of L:

$$l_{11} = a_{11} = 1$$

$$l_{21} = a_{21} = 2$$

$$l_{31} = a_{31} = 3$$

1st row of U:

$$uij = \frac{a_{ij}}{l_{11}}$$

$$u_{11} = 1$$

$$u_{12} = \frac{a_{12}}{l_{11}} = \frac{1+\varepsilon}{1} = 1 + 0.5 \cdot 10^{-15}$$

$$u_{13} = \frac{a_{13}}{l_{11}} = 3$$

2nd column of L:

$$l_{i2} = a_{i1} - l_{i1} \cdot u_{12}$$

$$l_{22} = a_{22} - l_{21} \cdot u_{12} = 2 - (2 \cdot (1+\varepsilon)) = -10^{-15}$$

$$l_{32} = a_{32} - l_{31} \cdot u_{12} = 6 - (3 \cdot (1+\varepsilon)) = 3 - 10^{-15}$$

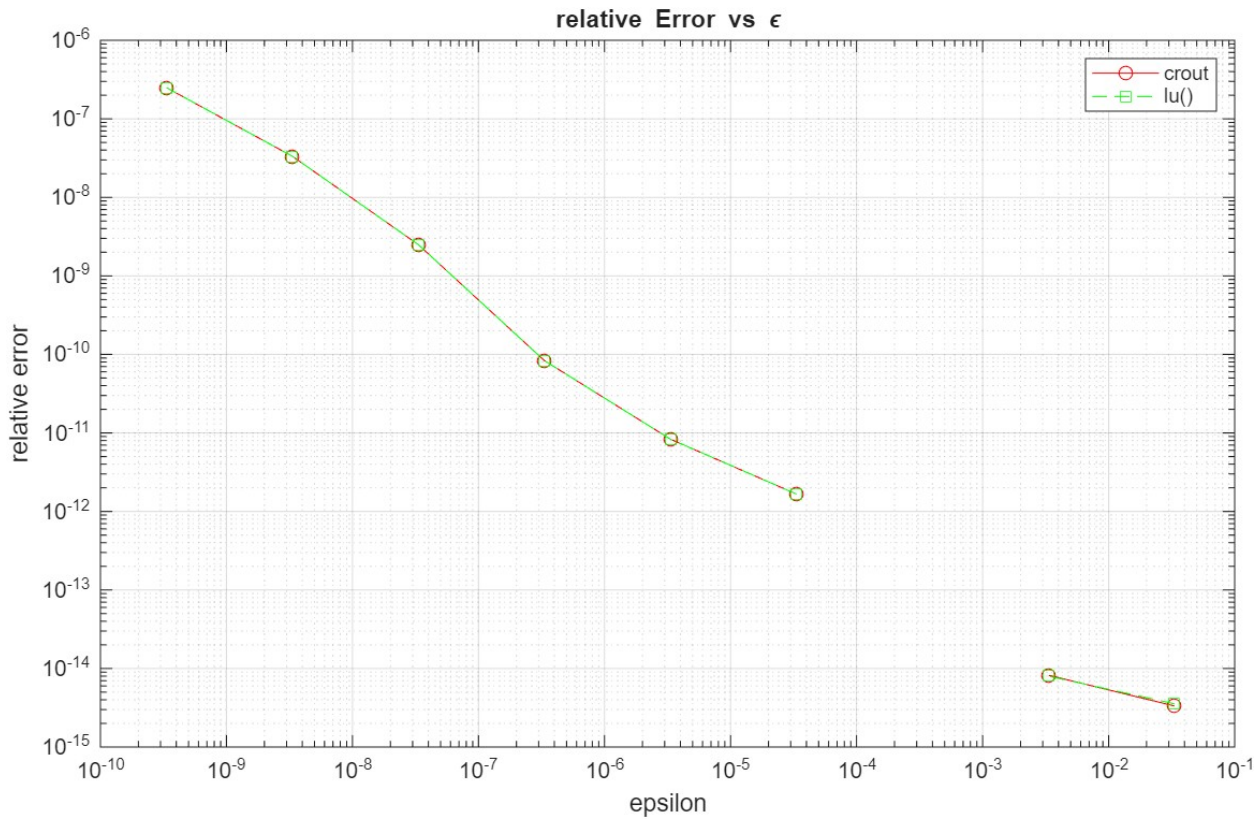


Figure 3: Enter Caption

2nd row of U:

$$u_{2j} = \frac{1}{l_{22}} \cdot (a_{2j} - l_{21} \cdot u_{1j})$$

$$u_{22} = 1$$

$$u_{23} = \frac{1}{l_{22}} \cdot (a_{23} - l_{21} \cdot u_{13}) = -1.4 \cdot 10^{16}$$

3rd column of L:

$$l_{33} = a_{33} - (l_{31} \cdot u_{13} + l_{32} \cdot u_{23}) = 4 - (3 \cdot 3 + (3 - 1.5 \cdot 10^{-15}) \cdot (-1.4 \cdot 10^{16}) \approx 4.2 \cdot 10^{16}$$

So our L and U matrices will look like this after decomposition:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & -2\varepsilon & 0 \\ 3 & 3 - 3\varepsilon & 4.2 \times 10^{16} \end{bmatrix}, \quad U = \begin{bmatrix} 1 & 1+\varepsilon & 3 \\ 0 & 1 & -1.4 \times 10^{16} \\ 0 & 0 & 1 \end{bmatrix}$$

To find the residual $R = A - LU$: We can automatically see that by calculating the product of LU, we run into an issue at the $lu_{33}$ position, which will be non-zero.

While computing the L matrix, we see that the $l_{22}$ element is very small ($-10^{-15}$), resulting in division with a number very close to 0. This creates a problem when calculating $u_{23}$ which becomes a huge number equal to $-1.4 \cdot 10^{16}$, resulting in round-off error when calculating $l_{33}$.

When we compute the residual of the Crout's method without pivoting, the residual has a maximum value of 4.0 in the $lu_{33}$ position. This signals that the round-off error took over our calculations and resulted in an unsuccessful attempt. Using the built-in method in Matlab we get $8.88e-16$ as the maximum value of the residual, which is a number really close to 0, proving that the decomposition was successful. This proves that pivoting is crucial to the accuracy of our decomposition.

# 6 Exercise 5

(a) It is evident that the decomposition is theoretically possible for every $\varepsilon \neq 0$. However, with the limits of our computational capabilities, we have to start computing the decomposition to see.

1st column of L:

$$l_{11} = a_{11} = 1$$

$$l_{21} = a_{21} = 2$$

$$l_{31} = a_{31} = 3$$

1st row of U:

$$uij = \frac{a_{ij}}{l_{11}}$$

$$u_{11} = 1$$

$$u_{12} = \frac{a_{12}}{l_{11}} = \frac{1-\varepsilon}{1} = 1 - \varepsilon$$

$$u_{13} = \frac{a_{13}}{l_{11}} = 3$$

2nd column of L:

$$l_{i2} = a_{i1} - l_{i1} \cdot u_{12}$$

$$l_{22} = a_{22} - l_{21} \cdot u_{12} = 2 - (2 \cdot (1-\varepsilon)) = 2\varepsilon$$

$$l_{32} = a_{32} - l_{31} \cdot u_{12} = 6 - (1 \cdot (1-\varepsilon)) = 3 + 3\varepsilon$$

2nd row of U:

$$u_{2j} = \frac{1}{l_{22}} \cdot (a_{2j} - l_{21} \cdot u_{1j})$$

$$u_{22} = 1$$

$$u_{23} = \frac{1}{l_{22}} \cdot (a_{23} - l_{21} \cdot u_{13}) = -\frac{2}{\varepsilon}$$

3rd column of L:

$$l_{33} = a_{33} - (l_{31} \cdot u_{13} + l_{32} \cdot u_{23}) = 1 + \frac{6}{\varepsilon}$$

It turns out that we only need to look at the pivot elements of L to avoid division by 0, so we have to make sure that $l_{11}, l22 and l_{33}$ are all non-zero to even perform the decomposition.

$$l_{11} = 1$$

, so we are safe to continue

$$l_{22} \neq 0 \implies 2\varepsilon \neq 0 \implies \varepsilon \neq 0$$

$$l_{33} \neq 0 \implies 1 + \frac{6}{\varepsilon} \neq 0 \implies 1 = \frac{-6}{\varepsilon} \neq \implies \varepsilon = -6$$

Therefore, LU decomposition without pivoting is not possible for $\varepsilon = 0$ and $\varepsilon = -6$. These are the specific values, however for any values of $\varepsilon$ close to 0, we would run into round-off error problems, just like we did in exercise number 4.

(b) A singular matrix has a determinant equal to 0. To find the $\varepsilon$ values, we calculate the determinant:

$$\det(A) = 1 \cdot \begin{vmatrix} 2 & 2 \\ 6 & 4 \end{vmatrix} - (1-e) \cdot \begin{vmatrix} 2 & 2 \\ 3 & 4 \end{vmatrix} + 3 \cdot \begin{vmatrix} 2 & 2 \\ 3 & 6 \end{vmatrix}$$

$$\det(A) = 1(8-12) - (1-e)(8-6) + 3(12-6)$$

$$\det(A) = -4 - 2(1-e) + 18 = 14 - 2 + 2e = 12 + 2e$$

$$12 + 2\varepsilon = 0 \implies e = -6$$

.

Let's evaluate: 1st col of L:

$$l_{11} = a_{11} = 1$$
$$l_{21} = a_{21} = 2$$
$$l_{31} = a_{31} = 3$$

1st row of U:

$$u_{11} = 1$$
$$u_{12} = a_{12}/l_{11} = 7/1 = 7$$
$$u_{13} = a_{13}/l_{11} = 3/1 = 3$$

2nd col of L:

$$l_{22} = a_{22} - l_{21}u_{12} = 2 - (2)(7) = -12$$
$$l_{32} = a_{32} - l_{31}u_{12} = 6 - (3)(7) = -15$$

2nd row of u:

$$u_{22} = 1$$
$$u_{23} = (a_{23} - l_{21}u_{13})/l_{22} = (2 - 2(3))/-12 = -4/-12 = 1/3$$

3rd col of L:

$$l_{33} = a_{33} - (l_{31}u_{13} + l_{32}u_{23}) = 4 - (3(3) + (-15)(1/3)) = 4 - (9 - 5) = 4 - 4 = 0$$

Therefore, the LU decomposition when $\varepsilon = -6$ is:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & -12 & 0 \\ 3 & -15 & 0 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & 7 & 3 \\ 0 & 1 & 1/3 \\ 0 & 0 & 1 \end{bmatrix}$$

(c)Since L is an upper triangular matrix, it's determinant is equal to the product of the diagonal elements. L has a zero as its last diagonal element, making the $det(L) = 0$.

Since $A = LU$, from linear algebra we know that the determinant of A should be the product of the determinants of L and U:

$$\det(A) = \det(L) \times \det(U)$$

Since $det(L) = 0 \implies det(A) = 0$, which means that A is a singular matrix.

# 7 Exercise 6

Prove that $K_2(A^2) = (K_2(A))^2$.

Since A is symmetric, we know that the $L^2$ natural norm is the absolute value of its largest eigenvalue and since it is positive symmetric, the eigenvalues are all positive. So, the $L^2$ norm of A is $\|A\|_2 = \lambda_{max}$. In order to find the condition number, we need to find the norm of the inverse of A. By definition, if $\lambda$ is an eigenvalue of A, then $\frac{1}{\lambda}$ is an eigenvalue of $A^{-1}$. Since we want the absolute maximum eigenvalue of the inverse, we need to find the smallest eigenvalue to make the fraction as large as we can. Therefore, the norm of $A^{-1}$:

$$\|A^{-1}\|_2 = \lambda_{max}(A^{-1}) = \frac{1}{\lambda_{min}(A)}$$

So our condition number is defined by:

$$K_2(A) = \|A\|_2 \cdot \|A^{-1}\|_2 = \frac{\lambda_{max}}{\lambda_{min}}$$

If $\lambda$ is an eigenvalue of A with the eigenvector u: $Av = \lambda v$

$$A^2 v = AAv = A(\lambda v) = \lambda(Av) = \lambda\lambda v = \lambda^2 v$$

So this means that the maximum/minimum eigenvalues of $A^2$ are the square maximum/minimum eigenvalues of A.

Thus,

$$K_2(A^2) = \frac{\lambda_{max}(A^2)}{\lambda_{min}(A^2)} = \frac{(\lambda_{max})^2}{(\lambda_{min})^2} = (\frac{\lambda_{max}}{\lambda_{min}})^2 = (K_2(A))^2$$

# 8 Exercise 7

Implementing Cholesky's method on matrix A, gives us :

$$L = \begin{bmatrix} 4 & 0 & 0 \\ -7 & 2 & 0 \\ 0 & 5 & 2 \end{bmatrix}$$

And for matrix B:

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 4 & 0 & 0 \\ 3 & 1 & 1 & 0 \\ -2 & 1 & 0 & 2 \end{bmatrix}$$

Both matrices are square symmetrical matrices and after running Cholesky's decomposition, we get the matrices above. Since Cholesky's decomposition was applied successfully, we can say that both matrices are symmetric, positive definite (see theorem below).

**Theorem 8.3.3**

*The following conditions are equivalent for a symmetric $n \times n$ matrix $A$:*

1. *$A$ is positive definite.*
2. $\det({}^{(r)}A) > 0$ *for each* $r = 1, 2, \ldots, n$.
3. $A = U^T U$ *where $U$ is an upper triangular matrix with positive entries on the main diagonal.*

*Furthermore, the factorization in (3) is unique (called the* ***Cholesky factorization***[6] *of $A$).*

Figure 4: Theorem